

[1] unemployment_analysis.ipynb

[2] ✓ 2s

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns
```

[5] ✓ 0s

```
df = pd.read_csv("Unemployment.csv")

df.head()
```

	DATE	SLUEM1524ZSIND
0	1991-01-01	11.792
1	1992-01-01	11.813
2	1993-01-01	11.835

[5] ✓ 0s

```
df = pd.read_csv("Unemployment.csv")
df.head()
```

	DATE	SLUEM1524ZSIND
0	1991-01-01	11.792
1	1992-01-01	11.813
2	1993-01-01	11.835
3	1994-01-01	11.845
4	1995-01-01	12.132

Next steps: [Generate code with df](#) [New interactive sheet](#)

[6] ✓ 0s

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
[6] ✓ 0s df.info()

... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 33 entries, 0 to 32
Data columns (total 2 columns):
#   Column                Non-Null Count  Dtype  
---  --
0   DATE                   33 non-null    object  
1   SLUEM1524ZSIND         33 non-null    float64  
dtypes: float64(1), object(1)
memory usage: 660.0+ bytes
```

```
[7] ✓ 0s df.isnull().sum()

0
DATE 0
SLUEM1524ZSIND 0
```

```
0 DATE 33 non-null object
1 SLUEM1524ZSIND 33 non-null float64
dtypes: float64(1), object(1)
memory usage: 660.0+ bytes

[7] ✓ 0s
df.isnull().sum()

0
DATE 0
SLUEM1524ZSIND 0
dtype: int64

[8] ✓ 0s
df.dropna(inplace=True)

[9] ✓ 0s
df.isnull().sum()

0
```



0s ut.dropna(inplace=True)

[9] 0s df.isnull().sum()

	0
DATE	0
SLUEM1524ZSIND	0

dtype: int64

[10] 0s df.describe()

	SLUEM1524ZSIND
count	33.000000
mean	17.248848
std	4.524133
min	11.700000


```
df.describe()
```

	SLUEM1524ZSIND
count	33.000000
mean	17.248848
std	4.524133
min	11.792000
25%	13.297000
50%	16.494000
75%	20.871000
max	25.939000

```
print(df.columns)
```

```
[11] print(df.columns)
Index(['DATE', 'SLUEM1524ZSIND'], dtype='object')

[14] df['DATE'] = pd.to_datetime(df['DATE'])

[15] df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 33 entries, 0 to 32
Data columns (total 2 columns):
#   Column          Non-Null Count  Dtype  
---  -
0    DATE             33 non-null    datetime64[ns]
1    SLUEM1524ZSIND    33 non-null    float64
dtypes: datetime64[ns](1), float64(1)
memory usage: 660.0 bytes
```

```
[15]
✓ 0s

df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 33 entries, 0 to 32
Data columns (total 2 columns):
#   Column          Non-Null Count  Dtype  
---  -
0   DATE             33 non-null    datetime64[ns]
1   SLUEM1524ZSIND   33 non-null    float64
dtypes: datetime64[ns](1), float64(1)
memory usage: 660.0 bytes
```

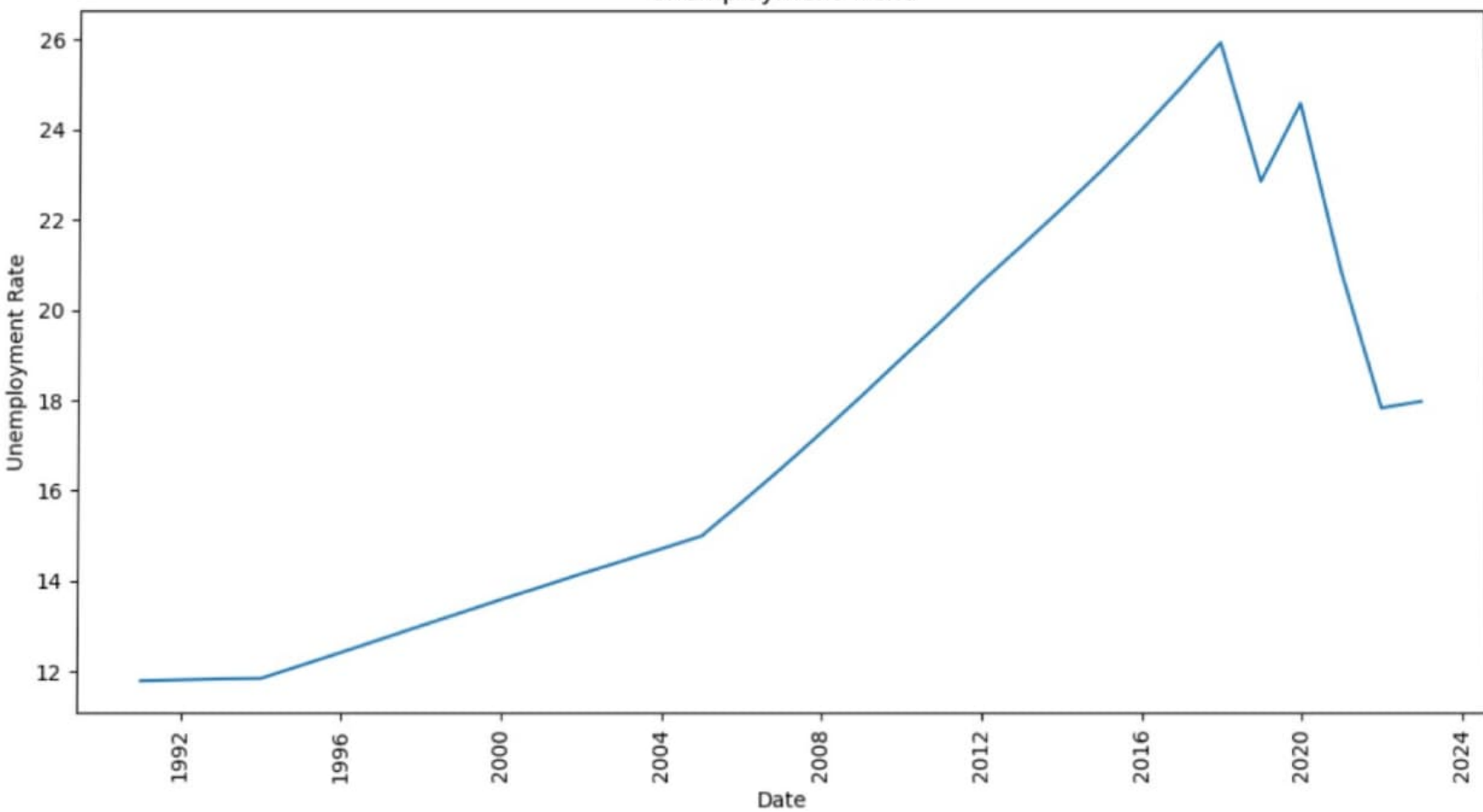
```
[17]
✓ 0s

plt.figure(figsize=(12,6))

plt.plot(
    df['DATE'],
    df['SLUEM1524ZSIND']
)

plt.title("Unemployment Trend")
plt.xlabel("Date")
```


Unemployment Trend



[] March 2020

[25] ✓ 0s
covid_data = df[
df['DATE'] >= '2020-03-01'
]

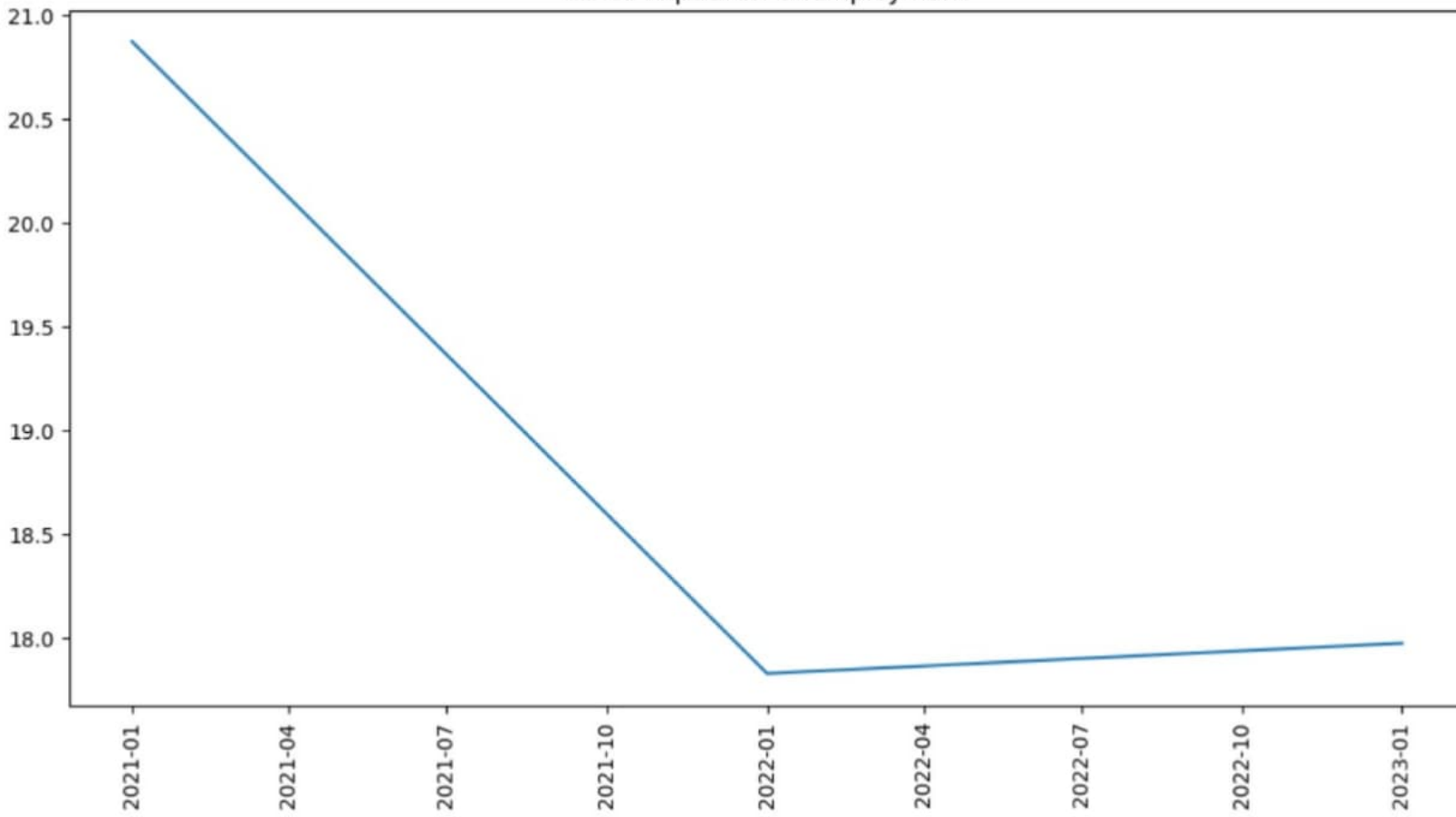
[20] ✓ 0s
covid_data.head()

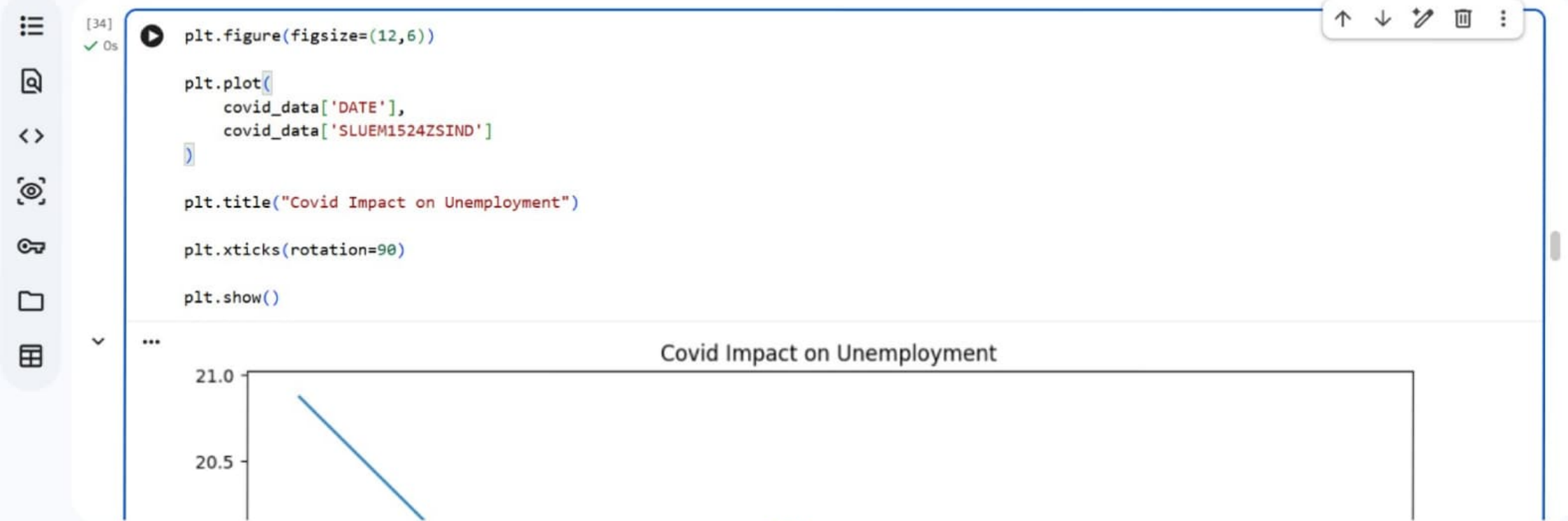
	DATE	SLUEM1524ZSIND
30	2021-01-01	20.871
31	2022-01-01	17.826
32	2023-01-01	17.972

Next steps: [Generate code with covid_data](#) [New interactive sheet](#)

[34] ✓ 0s
plt.figure(figsize=(12,6))

Covid Impact on Unemployment

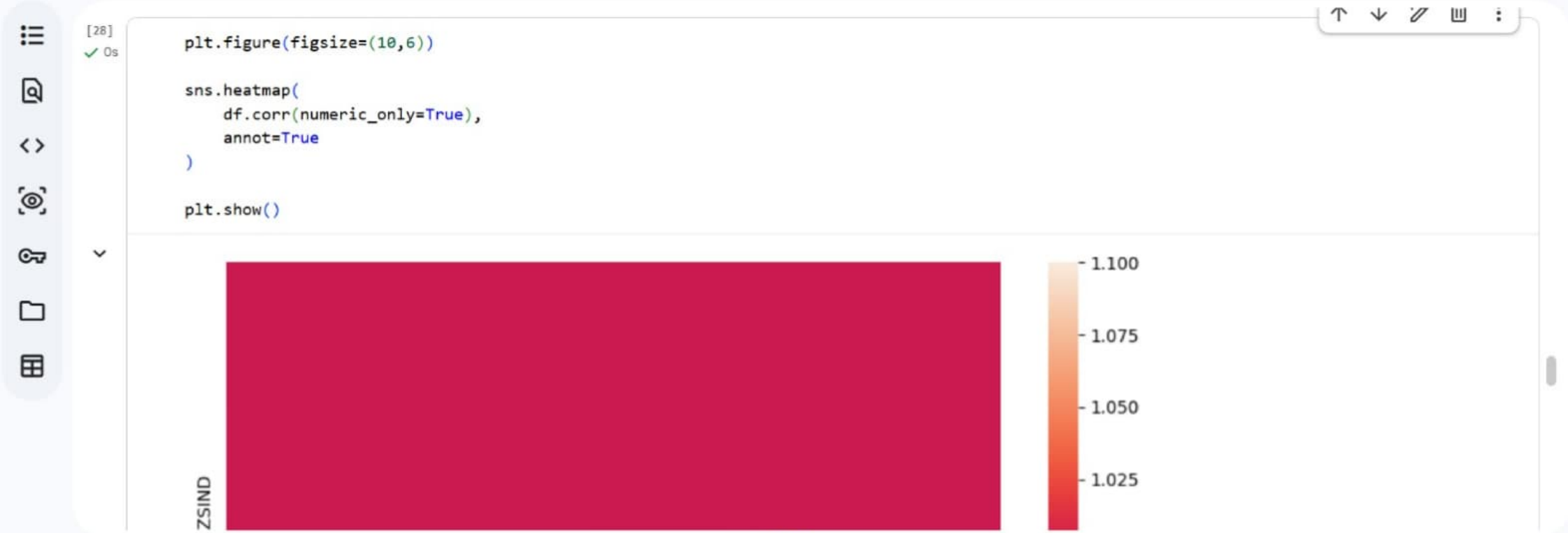




```
[36] ✓ 0s region_data = df.groupby('DATE')[ 'SLUEM1524ZSIND' ].mean()

print(region_data)
```

DATE	SLUEM1524ZSIND
1991-01-01	11.792
1992-01-01	11.813
1993-01-01	11.835
1994-01-01	11.845
1995-01-01	12.132
1996-01-01	12.418
1997-01-01	12.713
1998-01-01	13.007
1999-01-01	13.297
2000-01-01	13.589
2001-01-01	13.869
2002-01-01	14.157
2003-01-01	14.431





```
[30] df['Month'] = df['DATE'].dt.month
```

```
[31] df.head()
```

	DATE	SLUEM1524ZSIND	Month
0	1991-01-01	11.792	1
1	1992-01-01	11.813	1
2	1993-01-01	11.835	1
3	1994-01-01	11.845	1
4	1995-01-01	12.132	1

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
[42] monthly = df.groupby('Month')['SLUEM1524ZSIND'].mean()
```

```
print(monthly)
```



Next steps: [Generate code with df](#) [New interactive sheet](#)

```
[42]
✓ 0s
monthly = df.groupby('Month')['SLUEM1524ZSIND'].mean()

print(monthly)

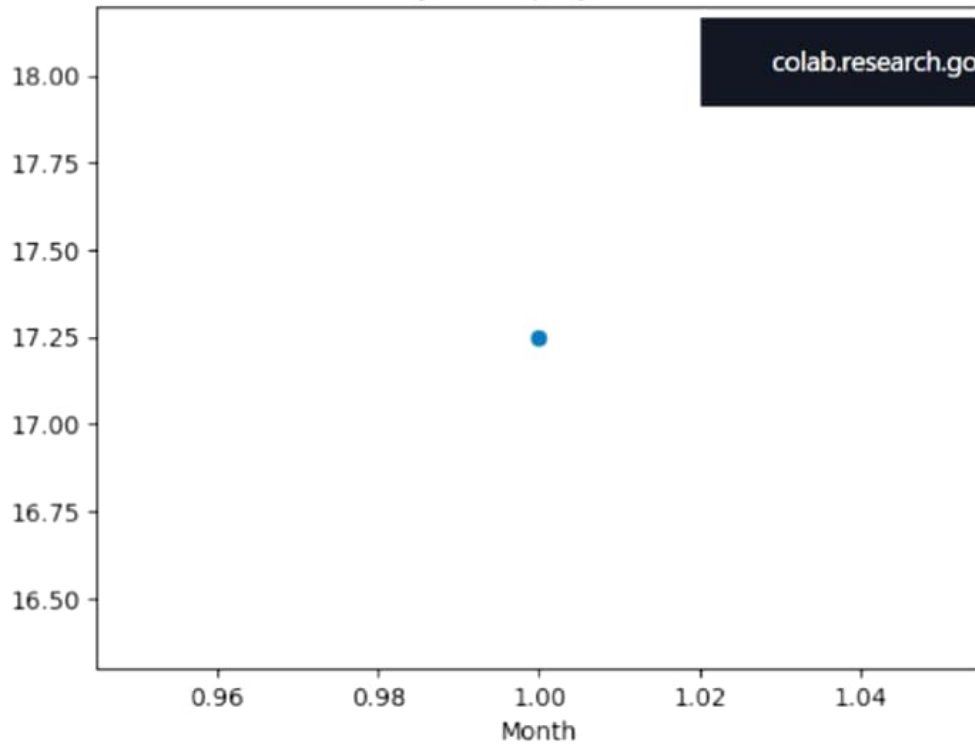
Month
1    17.248848
Name: SLUEM1524ZSIND, dtype: float64
```

```
[44]
✓ 0s
monthly.plot(
    kind='line',
    marker='o'
)

plt.title(
    "Monthly Unemployment Trend"
)

plt.show()
```

Monthly Unemployment Trend



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